

Q.1 Explain the concept of molecular taxonomy and its role in the five kingdom classification system.

Answer:

Molecular taxonomy is a branch of taxonomy that uses molecular data, such as DNA, RNA, and protein sequences, to classify and identify organisms. It provides a more accurate and objective method for understanding evolutionary relationships compared to traditional methods based on physical traits or morphology.

Concept of molecular taxonomy:

- It compares genetic material to study how closely organisms are related.
- Uses tools like DNA sequencing, ribosomal RNA analysis, and protein structure comparison.
- Helps in identifying species that may look similar but are genetically different.
- Useful for classifying microorganisms that have few distinguishing physical features.

Role in five kingdom classification system:

The five kingdom system includes:

- | | | |
|------------|-------------|----------|
| 1. Monera | 2. Protista | 3. Fungi |
| 4. Plantae | 5. Animalia | |

Molecular taxonomy has helped:

- Confirm genetic differences between kingdoms.
(eg. Prokaryotes vs. eukaryotes)
- Identify Archaea as genetically distinct from Bacteria, leading to the three domain system later.
- Improve the classification of protists, which are highly diverse and hard to group based on appearance.

Q.2 Discuss how model organisms such as *E. coli*, *S. cerevisiae*, *Drosophila melanogaster*, *Caenorhabditis elegans*, *Arabidopsis thaliana*, and *Mus musculus* represent different kingdoms and describe their significance.

Answer:

Organism	Kingdom	Significance.
<i>Escherichia coli</i> (<i>E. coli</i>)	Monera	A prokaryotic bacterium used in genetic engineering, gene cloning and recombinant DNA technology.
<i>Saccharomyces cerevisiae</i>	Fungi	A unicellular yeast used to study cell cycle regulation, DNA repair, and eukaryotic gene expression.
<i>Drosophila melanogaster</i>	Animalia	A fruit fly widely used in genetics, mutation analysis, and developmental biology.

<i>Caenorhabditis elegans</i>	Animalia	A transparent nematode (roundworm) used in studies of cell lineage, neural development, and apoptosis.
<i>Arabidopsis thaliana</i>	Plantae	A small flowering plant used as model for plant genetics, growth regulation, and photosynthesis studies.
<i>Mus musculus</i>	Animalia	A mouse used as a model for human disease, genetics functions, and drug testing due to its genetic similarity to humans.

Q.3 How many true-breeding pea plant varieties did Mendel select as pairs that were similar except one character with contrasting traits?

A. 2 B. 14 C. 6 D. 4

Answer:

B. 14

Q.4 Discuss the role of DICOM image formats and the concept of recombinant DNA technology.

Answers:

Role of DICOM Image Formats:

DICOM stands for Digital Imaging and Communications in Medicine.

Key Features are -

- It is a standard format for storing, transmitting and handling medical imaging data.
- Supports images from CT, MRI, ultrasound, X-Ray etc.
- Includes both the image data and patient information.

Importance in medicine -

- Ensures compatibility across different medical devices and software.
- Enables easy sharing of medical images across hospitals, clinics and doctors.
- Useful in diagnosis, treatment planning and medical record keeping.

Concept of Recombinant DNA Technology:

Recombinant DNA technology involves combining DNA from different sources to form a new genetic combination.

Key steps :-

1. Isolation of the gene of interest.
2. Cutting DNA using restriction enzymes.
3. Inserting the gene into a vector (like plasmid)
4. Transferring the recombinant vector into a host organism (like E. coli)
5. Expressing of the desired protein or gene in the host.

Applications :

- Production of insulin, growth hormones, vaccines.
- Development of genetically modified organisms (GMOs).
- Used in gene therapy and medical diagnostic.

Q.5 Analyze the significance of the DNA Technology (use and application) Regulation Bill, 2019.

Answer:

The DNA Technology (Use and Application) Regulation Bill, 2019 was introduced in the Indian parliament to regulate the use of DNA technology for identification of individuals in various situations, such as criminal investigations, civil cases, and missing persons.

Key features of the Bill :

1. Regulation of DNA profiling for criminal and civil matters
2. Establishment of DNA Regulatory Board to oversee and ensure ethical use of DNA data.
3. Creation of national and Regional DNA data banks to store DNA profiles.
4. Lays out procedures for collection, analysis and storage of DNA samples.
5. Includes provisions for privacy protection, misuse prevention, and penalties for violations.

Significance of the Bill :

- Strengthens forensic investigations : Helps in identifying criminals, victims, missing persons, and disaster victims.
- Supports justice system : Reduces wrongful convictions and improves the accuracy of evidence.
- Enhances national security : Enables rapid identification in cases of terrorism and major crimes.
- Protects individual privacy : Ensures DNA is collected and used ethically and legally.
- Boosts scientific credibility : Provides a legal framework for DNA labs to function under strict standards.